A PBL REPORT ON

SMARTEXIT: AI POWERED FRICTIONLESS CHECKOUT SYSTEM

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CERTIFICATE

This is to certify that the project report entitled

"SMARTEXIT: AI POWERED FRICTIONLESS CHEKOUT SYSTEM."

is a bonafide work carried out by him/her under the supervision of **Prof. R. P. Pawar** and it is approved for the partial fulfillment of the requirements of Savitribai Phule Pune University, Pune for the award of the degree of Bachelor of Engineering (Computer Engineering) during the year **2025-26**.

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ABSTRACT

Long queues and slow billing counters continue to frustrate customers in physical retail stores, especially in a world where speed and convenience are increasingly expected. Traditional checkout systems not only create delays but also limit the overall shopping experience. To address this, SmartExit introduces a frictionless checkout approach that empowers customers to complete their purchases quickly and independently, without needing to wait in line or interact with a cashier.

SmartExit allows users to scan items using their smartphones or smart carts, view real-time pricing with tax breakdowns, and pay instantly through digital methods such as UPI, cards, or mobile wallets. Once payment is complete, an e-receipt and a secure exit QR code are generated for verification by store personnel or via automated gates equipped with RFID or NFC technologies. The system is designed to be fast, reliable, and user-friendly, significantly reducing wait times and improving customer satisfaction.

The platform is developed using a Flutter mobile application with backend support provided by Firebase, enabling secure authentication, real-time data synchronization, and cloud-based storage through Firestore. It seamlessly integrates with trusted payment gateways such as Razorpay, Stripe, or Paytm to facilitate instant and secure digital transactions. Additional features like barcode scanning, smart cart functionality, and contactless verification enhance the system's efficiency and user experience. Designed for flexibility and scalability, SmartExit provides a modern retail solution that combines intuitive design with advanced technology—transforming the traditional checkout process into a faster, simpler, and more seamless experience aligned with the expectations of today's consumers.

Keywords: Frictionless Checkout, SmartExit, Self-Checkout, Digital Payments, Retail Automation, Barcode Scanning, RFID, NFC, Mobile App, React Native, FastAPI, Customer Experience.

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INTRODUCTION

1.1 Background and Basics

With the growing demand for convenience and efficiency in the retail industry, traditional checkout systems have become a major source of customer dissatisfaction due to long queues and slow billing processes. As consumer expectations shift toward faster and contactless experiences, retailers must adopt smarter solutions that enhance speed, accuracy, and customer satisfaction.

SmartExit addresses these challenges by introducing an automated and frictionless checkout system that allows customers to complete their purchases independently, without relying on cashiers or waiting in line. Users can scan items using their smartphones or smart carts, view real-time pricing with tax breakdowns, and make instant digital payments through UPI, cards, or mobile wallets. After successful payment, the system generates an e-receipt and a secure exit QR code, which can be verified by store personnel or through RFID/NFC-enabled automated gates.

The platform is built using a Flutter mobile application for the frontend, supported by a Node.js or FastAPI backend for efficient processing and integration with trusted payment gateways such as Razorpay, Stripe, or Paytm. With additional features like barcode scanning, smart carts, and contactless verification, SmartExit provides a scalable, secure, and user-friendly solution for modern retail environments.
By combining intuitive design with cutting-edge technology, SmartExit redefines the retail checkout experience making it faster, simpler, and more aligned with the digital-first expectations of today's consumers.

1.2 Literature Survey

Several research works have been conducted in the area of automated web vulnerability scanning, focusing on improving accuracy, scalability, and usability of scanning tools. This section presents a review of key existing systems and their contributions.

1. Secure Digital Wallet by Using QR Code: Mapping of Mobile Payment in Partially Connected Environment

Khan, M. A., Raza, A., Ahsan, K., and Irshad, S. (2024) say that mobile payment systems can be significantly improved using QR code technology, even in environments with limited or unstable internet connectivity. Their study discusses how QR codes can be effectively integrated into digital wallets to enhance transaction security, reliability, and accessibility. The authors highlight that such systems can help overcome traditional payment issues related to connectivity and manual verification. This work is relevant to the *Smart Exit* project as it supports the concept of secure and fast digital transactions, an essential component in frictionless checkout systems where instant payments and real-time verification are critical.

2. Smartcart Vision

Tembe, S., Thakur, S., Patil, P., Bhoir, A., Madrewar, K., and Jadhav, D. say that the Smartcart Vision model aims to simplify the retail checkout process by utilising computer vision technology to automatically detect and identify products in a shopping cart. Their approach removes the need for manual barcode scanning, significantly reducing human intervention and checkout time. The system uses cameras and trained machine learning models to recognise items visually, improving accuracy and convenience. This research directly relates to the *Smart Exit* system as both share the common goal of automating retail checkout processes through vision-based AI, thus enhancing efficiency and reducing customer waiting time.

3. A Computer Vision-based Model to Implement Automation in Retail Checkouts Using YOLOv5

Jain, A., Kumar, K., Yadav, N., and Chaubey, K. B. say that automation in retail checkouts can be achieved effectively using computer vision models like YOLOv5, which can detect multiple objects in real time. Their system proposes the use of pretrained YOLOv5 models to identify products in shopping carts and automate billing processes, eliminating the need for manual scanning. The authors point out challenges such as occlusion, overlapping products, lighting variations, and the requirement for product-specific datasets to maintain detection accuracy. Their findings are particularly important for *Smart Exit*, as they emphasise the technical and data-related considerations that must be addressed when developing a robust, real-time AI-powered checkout system.

4. Systematic Review of Cashierless Stores (Just Walk Out Stores) Revolutionizing the Retail
Szabó-Szentgróti, E., Rámháp, S., and Kézai, P. K. (2023) say that cashierless store technology, often referred to as "Just Walk Out" systems, is revolutionising the retail sector through the integration of AI, computer vision, and sensor-based tracking. Their systematic review analyses how major innovations—such as those used by Amazon Go—enable customers to shop freely and leave without manual billing, as transactions are processed automatically through AI-based systems. The paper discusses the economic impact, technological advancements, and future challenges of implementing such systems at scale. This study provides a strong theoretical foundation for *Smart Exit*, supporting its objective to deliver a frictionless, contactless, and intelligent retail checkout experience.

1.3 Project Undertaken

1.3.1 Project Statement

With the growing demand for speed, convenience, and contactless experiences in the retail industry, traditional checkout systems have become a major bottleneck, resulting in long queues, billing delays, and reduced customer satisfaction. Many retail stores still rely on manual or semi-automated billing methods that lack real-time synchronization, digital payment support, and efficient exit verification mechanisms.

The proposed project, SmartExit – Frictionless Checkout System, aims to overcome these challenges by introducing an automated and intelligent retail checkout solution that allows customers to complete purchases independently—without waiting in line or interacting with a cashier. The system enables users to scan items using smartphones or smart carts, view real-time pricing with tax details, and make instant digital payments through UPI, debit/credit cards, or mobile wallets. Once payment is successful, an e-receipt and a secure exit QR code are generated for verification by store personnel or through RFID/NFC-enabled gates.

The application utilizes a Flutter-based mobile frontend for cross-platform accessibility, with Firebase serving as the backend for authentication, real-time data synchronization, and secure storage through Cloud Firestore. It also integrates payment gateways such as Razorpay, Stripe, or Paytm to enable seamless and secure transactions.

This project aims to deliver a user-friendly, scalable, and efficient retail checkout solution that minimizes billing time, enhances operational efficiency, and aligns with the expectations of modern, technology-driven consumers.

1.3.2 Statement of Scope

The scope of this project includes the design, development, integration, and deployment of a complete web-based vulnerability scanning system with the following capabilities:

- User Authentication & Authorization:
  - Secure user registration and login using JWT-based authentication
- Scanning Functionalities:
  - **Product Scanning:** Users can scan items using smartphones or smart carts through barcode or QR scanning
  - **Real-Time Billing:** Automatic price calculation with tax breakdown and total amount display
  - **Digital Payment Integration:** Support for multiple payment methods including UPI, debit/credit cards, and mobile wallets
  - **E-Receipt Generation:** Automatic generation of digital receipts upon successful payment
  - **Secure Exit Verification:** Generation of unique exit QR codes for authentication at store gates or verification via RFID/NFC systems
- User & Admin Dashboards:
  - **Customer Dashboard:** Allows users to view cart details, transaction history, and e-receipts
  - Real-time monitoring of payment and checkout activities
- Backend & Database Integration:
  - **Backend:** Backend services powered by Firebase, leveraging Firestore for secure data storage and synchronization.
  - Firebase Cloud Functions for server-side logic and payment verification
  - **Payment Gateway Integration:** Incorporation of trusted gateways such as Razorpay, Stripe, or Paytm for secure transactions
- Frontend Interface:
  - Built with Flutter for responsiveness

    - Intuitive, responsive, and user-friendly interface for smooth navigation and interaction
- Deployment & Demonstration:
    - The system can be deployed on a local or cloud environment
    - Ready for demonstration as a BE final year project

This project focuses on delivering a smart, secure, and efficient checkout experience through automation and digital payment integration. While it does not include advanced inventory management or AI-based product recognition, its modular architecture allows for future extensions such as facial recognition checkout, loyalty integration, or predictive analytics for enhanced customer engagement.

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2 PROJECT PLANNING AND MANAGEMENT

2.1 Detailed System Requirement Specification (SRS)

2.1.1 System Overview

The proposed system is a web-based automated checkout application that enables users to complete their purchases independently through an integrated scanning and digital payment system. The application eliminates the need for manual billing counters, providing a faster, contactless, and more efficient shopping experience for customers.

The application is designed with a Flutter frontend for an interactive and responsive user interface, Firebase Cloud Functions as the backend for processing and logic handling, Firestore for secure and real-time data storage, and integration with trusted payment gateways such as Razorpay, Stripe, and Paytm for digital transactions.

The system supports two user roles:

- **User:** Can register, log in, scan items using smartphones or smart carts, view real-time billing details, make digital payments, and access e-receipts and transaction history.
- **Admin:** Has additional privileges to monitor all checkout activities, manage users, verify payments, and oversee overall system performance.

The system performs multiple automated operations such as:

- **Product Scanning:** Barcode or QR-based product identification through smartphones or smart carts.
- **Real-Time Billing:** Automatic calculation of item prices with tax breakdown and total amount display.
- **Digital Payment Integration:** Secure payment handling via UPI, debit/credit cards, or mobile wallets.
- **E-Receipt Generation:** Automatic generation of digital receipts upon successful payment.
- **Secure Exit Verification:** Creation of unique exit QR codes verified by store staff or RFID/NFC-enabled gates.
- **Data Management:** Real-time synchronization of user activity, payment details, and scan records through Firestore.

By combining automation, cloud integration, and digital payment systems, SmartExit delivers a secure, scalable, and user-friendly solution for retail environments — enhancing both customer convenience and store efficiency.

2.1.2 Functional Requirements

The functional requirements define what the SmartExit system should do.

1. User Authentication & Authorization

- Users should be able to register and log in securely using JWT-based authentication.
- The system should differentiate between user roles (User/Admin).
- User passwords must be encrypted before being stored in the database.
- Admins should have privileges to manage user accounts and update roles.

2. Product Scanning & Billing

- Users can scan products using smartphones or smart carts via barcode or QR code.
- The system should retrieve product details (name, price, tax) from the database in real time.

- The application should automatically calculate total amounts and taxes during checkout.
- The system should handle multiple products in a single transaction efficiently.

3. Digital Payment Processing

- The system should support multiple payment methods, including UPI, debit/credit cards, and mobile wallets.
- Integration with secure payment gateways (Razorpay, Stripe, Paytm) must be implemented.
- Payment verification should be handled by Firebase Cloud Functions for security.
- After successful payment, an e-receipt must be generated and stored in the database.

4. Secure Exit Verification

- The system should generate a unique exit QR code after every successful transaction.
- The exit QR should be verified at the store gate by staff or RFID/NFC-enabled systems.
- Invalid or reused QR codes should be automatically rejected by the system.

5. User Dashboard

- Display the cart summary, payment history, and e-receipts of the logged-in user.
- Allow users to initiate new shopping sessions and view completed transactions.
- Provide real-time updates on billing and payment status.

6. Admin Dashboard

- View and manage all user accounts (update roles, deactivate accounts, view activity).
- Monitor real-time transactions, payments, and checkout activity.
- Access and analyze system-wide statistics such as daily sales and number of active users.
- Ensure secure and authorized access to all administrative functions.

2.1.3 Non-Functional Requirements

Security

- All communication between frontend and backend must be secured using HTTPS.
- JWT-based authentication should protect all API endpoints.
- Role-based authorization must be enforced for admin operations and payment handling.
- User passwords and sensitive data must be encrypted before storage.
- Integration with trusted and verified payment gateways (Razorpay, Stripe, Paytm) must ensure secure transactions.

• Usability

- The user interface should be simple, intuitive, and responsive, ensuring smooth navigation.
- Display clear success and error messages for all user actions.
- Maintain consistent design and branding throughout the application.
- Provide a seamless checkout experience across all devices.

• Scalability

- The system should support multiple concurrent users and transactions without performance issues.
- Backend services (Firebase Cloud Functions) and Firestore must be scalable to handle growing data volume.

- The architecture should allow future integration of features like loyalty programs or AI-based checkout.

• **Maintainability**

- The codebase should follow a modular and component-based architecture to simplify updates and debugging.
- Firebase functions, payment modules, and UI components should be independent and reusable.
- The system should be easily upgradable to incorporate new payment APIs or technologies.

• **Performance**

- The application should **load quickly** and respond to user actions with minimal delay.
- **Real-time billing and updates** must reflect instantly during product scanning.
- Payment processing and verification should complete within **a few seconds** under normal conditions.
- Optimize queries and network calls to **reduce latency and resource usage.**

• **Reliability**

- The system should handle **invalid inputs and failed transactions gracefully**.
- Implement **robust error handling and logging** for debugging and monitoring.
- Ensure **data consistency** even in case of network interruptions or device disconnection.
- Provide **fail-safe mechanisms** for payment retry and QR code regeneration if needed.

2.1.4 Deployment Environment

Component	Technology / Tools
Frontend	Flutter
Backend	Firebase Cloud Functions
Database	Firestore
Authentication	Firebase Authentication (JWT-based)
Hosting (Optional)	Local server / Cloud (e.g., Render, Railway, or AWS)
OS	Windows / macOS / Linux
Development Tools	Android Studio, VS Code, Postman, Draw.io, GitHub

2.1.5 External Interface Requirements

1. User Interfaces:

- Mobile application interface built using **Flutter**.
- Screens include **Login/Signup**, **QR/Barcode Scan Page**, **Dashboard**, and **Payment Summary**.
- **Admin Panel** (optional web interface) for managing users, transactions, and system logs.

2. Software Interfaces:

- **Firebase SDKs** for real-time communication between app and backend.
- **Cloud Functions** handle server-side logic and integration with external APIs (e.g., payment gateway).
- **Firestore Database** for data storage and retrieval.
- **Python-based Scanner Module** integrated through a secure API or SDK bridge.

3. Hardware Interfaces:

- Standard **smartphone camera** for QR/Barcode scanning.
- **Android or iOS device** with internet connectivity.
- **Computer/laptop** for admin access and system monitoring (optional).

4. Communication Interfaces:

- **HTTPS** protocol for secure data exchange between app and Firebase.
- **Firebase Cloud Messaging (FCM)** for notifications and alerts.
- **API calls** for payment processing and data synchronization.

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2.1.6 Other Requirements (Software Quality Attributes)

- **Interoperability:**
  The system can integrate with third-party services such as payment gateways, cloud analytics tools, or IoT-based parking sensors in the future.
- **Portability:**
  The application can run on both **Android** and **iOS** platforms, with optional web access for admin using Firebase's cross-platform support.
- **Extensibility:**
  New features such as **AI-based parking predictions**, **automated gate control**, or **extended analytics modules** can be easily added without major architectural changes.
- **Modularity:**
  The system follows a modular design separating **frontend (Flutter app)**, **backend logic (Firebase Functions/Python)**, and **database (Firestore)** to ensure maintainability and scalability.
- **Testability:**
  The app components, APIs, and Firebase functions are designed for **unit and integration testing**, ensuring stable and reliable performance.

2.2 Cost & Efforts Estimates

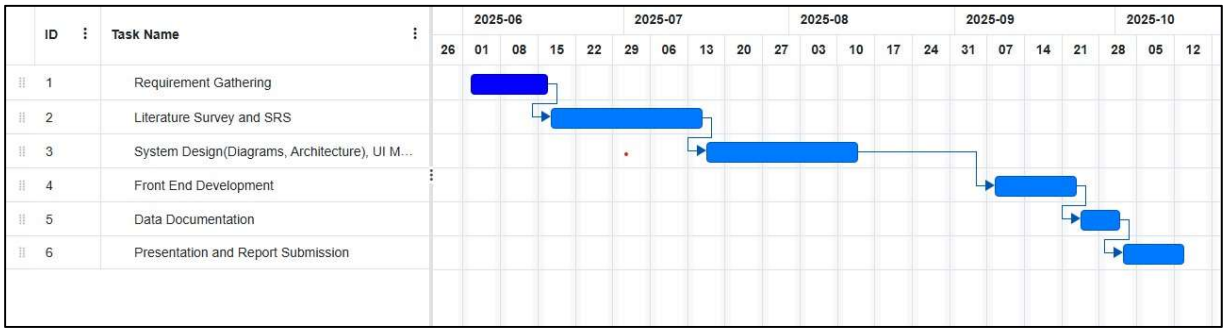
Resource	Estimated Cost
Development Tools (Open-source)	₹0 (Free: Flutter, Firebase, Python)
Hosting (Optional Cloud)	₹0 – ₹1000/month (free tiers available)
Developer Time	2 team members × 2 months × 80 hrs = 320 hrs
Documentation & Reports	₹0 (prepared in-house)

Total Estimated Cost: ₹0 – ₹2000 (if using free hosting tiers)
Effort Estimate: ~320 hours (including design, development, testing, and documentation)

2.3 Project Scheduling

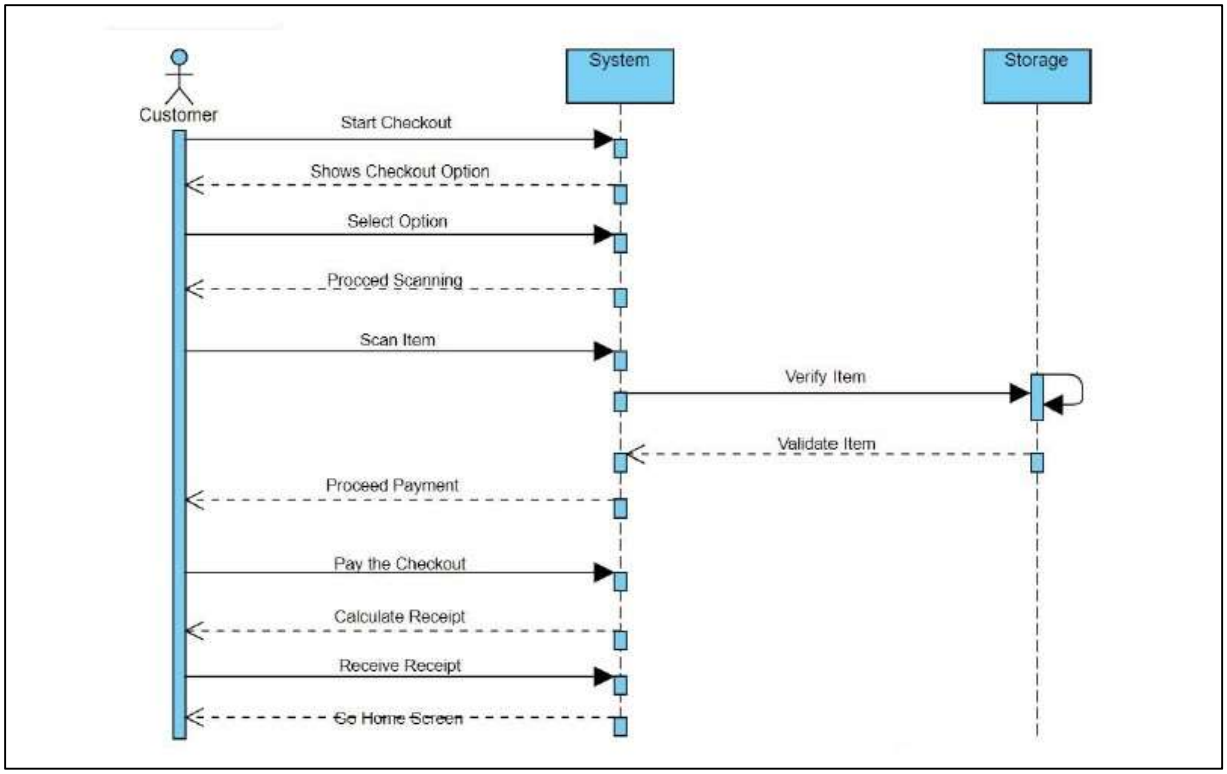
The project was planned using **incremental development** with clear milestones. The timeline was divided into **four major phases**: Requirements, Design, Development, and Testing & Deployment.

2.3.1 Timeline Chart

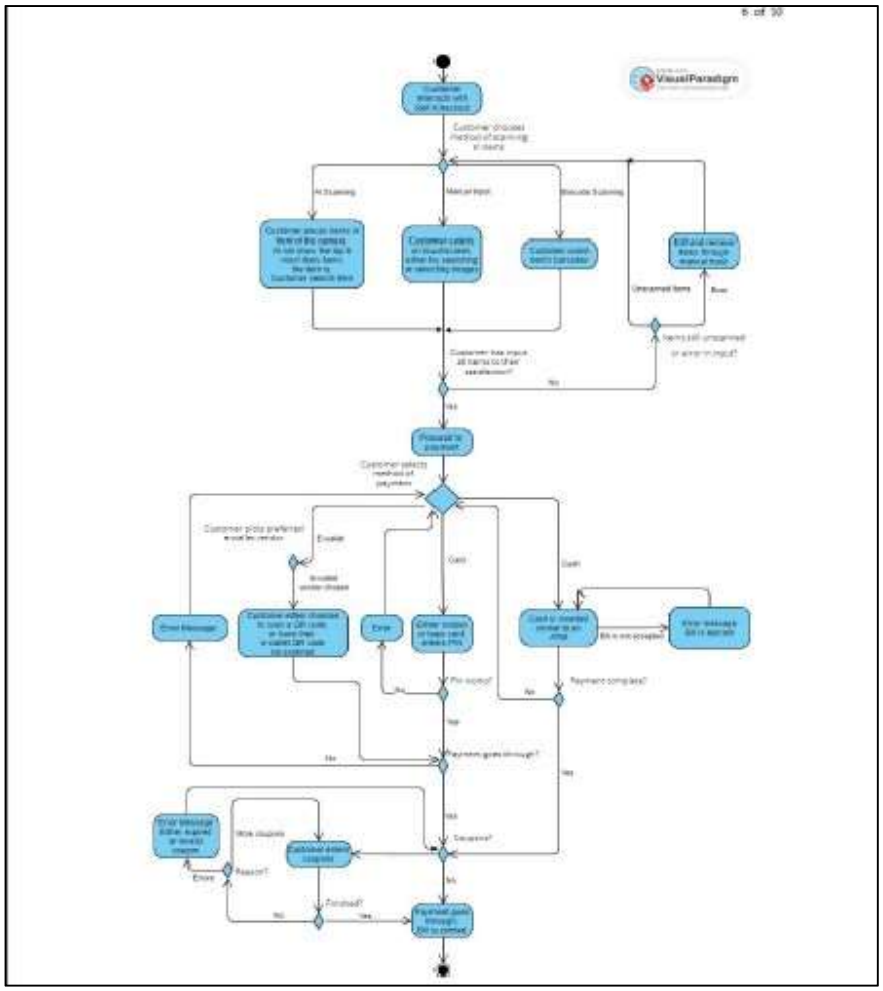


3.Analysis and Design

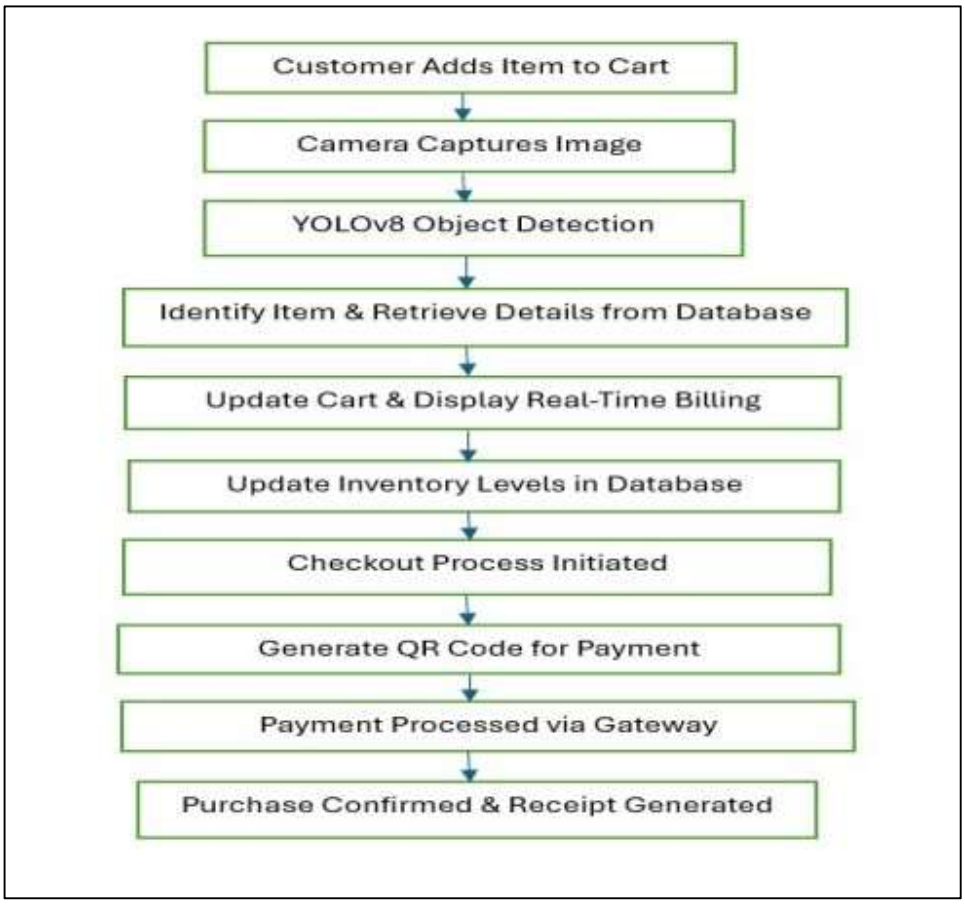
3.1 Sequence Diagram :



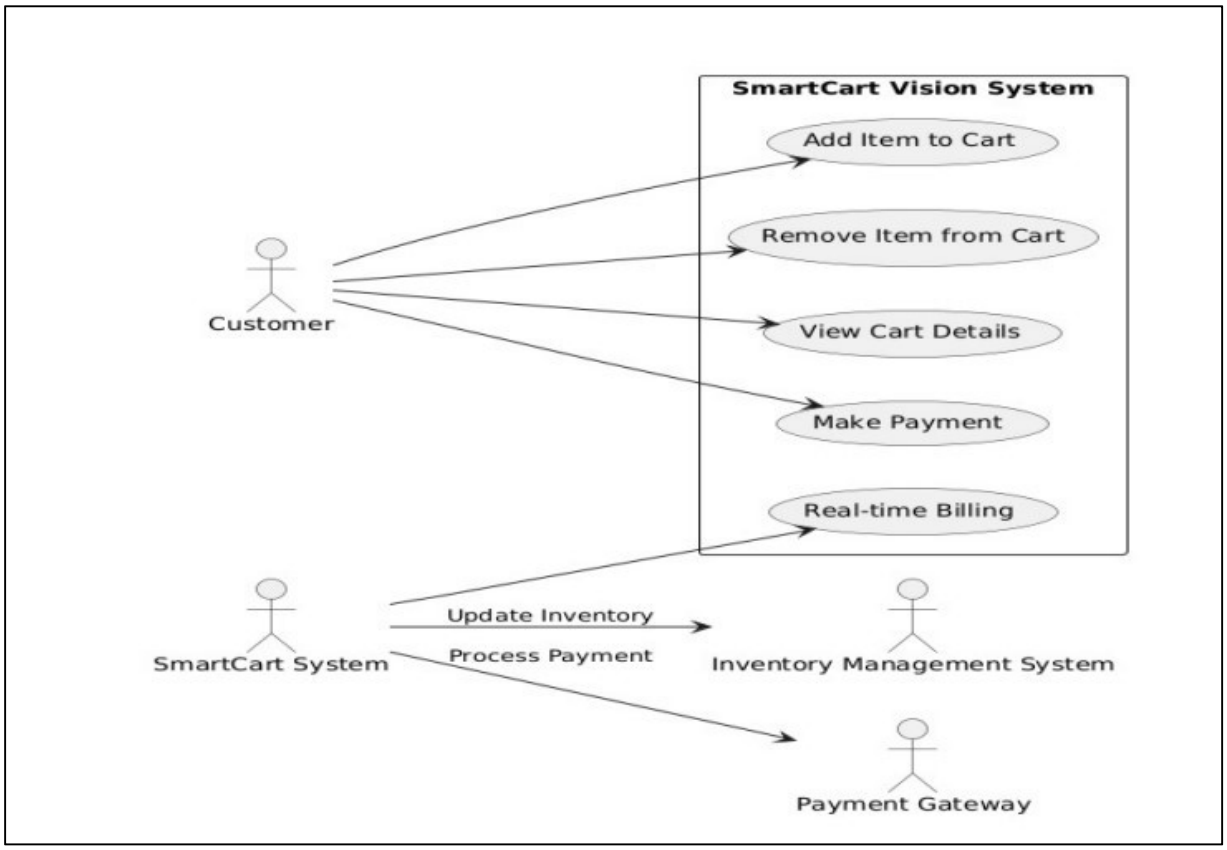
3.2 Activity Diagram :



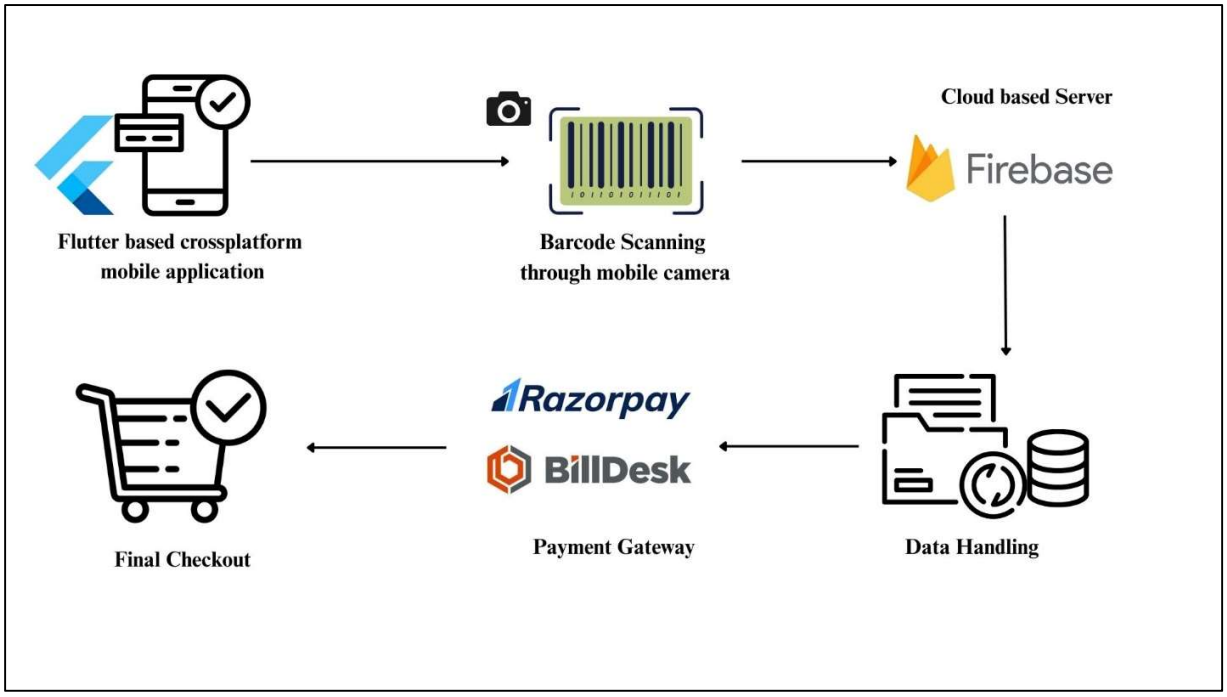
3.3 DataFlow Diagram :



3.4 UseCase Diagram :

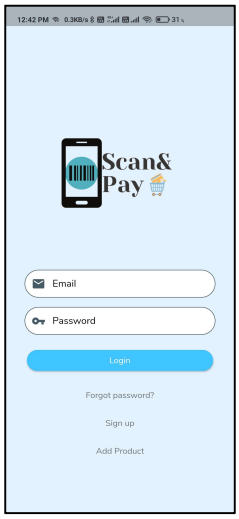


3.5 System Architecture



3.6 Result Analysis

Home Page



Login Page

4. Future Work

While the current system successfully demonstrates automated, contactless checkout using AI and Firebase integration, several enhancements can be explored in future development.

1. **Integration with Computer Vision Models:** Incorporating advanced object detection algorithms such as YOLOv8 or EfficientDet can improve product recognition accuracy, especially in cases of occlusion or overlapping items.
2. **Dynamic Pricing and Offer Recommendations:** Future versions can include AI-driven personalized offers, loyalty programmers, and dynamic pricing models to enhance the shopping experience and promote customer engagement.
3. **Offline Transaction Support:** Implementing offline data caching and sync mechanisms would allow transactions to continue even in areas with unstable internet connectivity, ensuring reliability.
4. **Enhanced Security and Fraud Detection:** Machine learning models can be introduced to detect fraudulent behavior, unauthorized payments, or inventory manipulation, strengthening system security.
5. **Scalability for Large Retail Chains:** The system can be optimized for multi-store management, enabling centralized control of inventory, pricing, and analytics for large-scale retail operations.
6. **User Analytics and Feedback System:** A feedback and analytics module could help store managers gain insights into customer preferences, shopping patterns, and system performance, supporting data-driven decision-making.

In conclusion, future work on Smart Exit will focus on improving **accuracy, security, scalability, and user experience**, ultimately creating a robust and intelligent solution for the next generation of retail automation.

5. Conclusion

In conclusion, Smart Exit: AI-Powered Frictionless Checkout System marks a significant step forward in modernizing the retail shopping experience through the integration of artificial intelligence, cloud computing, and intuitive interface design. The system has been designed to address one of the most common challenges in retail — long queues and time-consuming manual checkouts — by offering a fully automated, contactless, and efficient solution.

Developed using JavaFX, Smart Exit provides an interactive and user-friendly interface for both customers and store managers. The use of Firebase as the backend ensures secure and real-time management of user data, authentication, inventory, and transactions. The incorporation of AI algorithms enables the system to automatically detect products, calculate total amounts, and process payments without the need for human intervention, thus ensuring speed, accuracy, and convenience.

Overall, Smart Exit provides a robust foundation for the next generation of frictionless retail systems. Future improvements such as advanced computer vision models, scalable multi-store integration, enhanced fraud detection, and offline transaction capabilities could make the system even more powerful and adaptable. By continuing to refine and expand upon this work, Smart Exit has the potential to revolutionize retail operations and set new standards for speed, security, and customer convenience in the shopping industry.

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